Maxime SKRYMA

+33 6 18 24 35 40 | maxime.skryma@gmail.com | LinkedIn | GitHub

EDUCATION

ENSAE Paris **Paris, France**
Grande École Engineering Program *Expected 2028*
- GPA: 4.0/4.0 | Coursework: Machine Learning, Advanced Statistics, Monte Carlo Methods, Stochastic Calculus

HEC Paris **Paris, France**
Grande École / Master in Management *Expected 2028*
- GPA: 3.8/4.0 | Coursework: Data Science, Decision & Data Modelling, Statistics, Corporate Finance

RESEARCH EXPERIENCE

Research Intern, MathFi Lab – CMAP, École Polytechnique **Palaiseau, France**
Multi-agent reinforcement learning – Supervised by Prof. Huyên Pham *June–November 2026*
- Co-authored a paper submitted to ICLR 2027 on Q-learning for Mean-Field Control with heterogeneous, continuously-indexed agents, deriving convergence and sample-complexity guarantees
- Engineered AI-agent workflows that autonomously design simulations, run experiments and iterate on results, over 1,000+ CPU-hours
- Implemented the algorithm and deep RL baselines (DDPG, TD3) in PyTorch (code released with the paper), validated on optimal execution, systemic risk and network congestion control

AI & ENGINEERING PROJECTS

Crypto Market Making with Hawkes Processes & Deep RL (personal project) *2026*
- Built an end-to-end pipeline on Bitcoin tick data, including a multivariate Hawkes model of order flow coded from scratch (MLE with $O(N)$ recursion, simulation, goodness-of-fit tests)
- Formulated market making as an event-driven semi-Markov decision process and derived its HJB and Bellman equations, providing exact solutions to validate learned policies
- Trained deep RL agents (TD3, SAC) in the calibrated simulator, validated them against exact solutions and evaluated them on historical L2 order-book data against an analytical baseline

Atelier – Orchestration of AI Coding-Agent Teams (personal project) *2026*
- Built a framework orchestrating configurable teams of AI coding agents (Claude Code, Codex, Gemini) through their CLIs; runs on a VM or in Docker

Machine Learning Research Project in collaboration with Ekimetrics *2025–2026*
- Built a two-stage reliability framework: a secondary XGBoost model predicts a classifier's errors, with SHAP
- Built conformal prediction sets with finite-sample coverage guarantees to flag unreliable predictions for human review, evaluated on healthcare and banking datasets

PROFESSIONAL EXPERIENCE

TotalEnergies **Paris, France**
M&A Analyst Intern – Gas, Renewables & Power *June–July 2025*
- Worked on an €800M+ cross-border acquisition: valuation, deal materials, screening of 300+ comparables
- Presented quantitative market analyses to the CTO and Head of M&A, translating findings for executives

Square Capital **Paris, France**
Portfolio Analyst Intern *July–August 2024*
- Analysed investment cases, valuations, peer benchmarks and structured products for a $105M equity portfolio

SKILLS & INTERESTS

ML & Programming: Python (PyTorch, scikit-learn, XGBoost, pandas), NLP (text classification), Git, Linux
Languages: French (native), English (fluent), Spanish (B2), Russian (B1)
Associations: HEC Bourse (Head of Partnerships), Duo Conseil X-HEC, HEC Investment Club, Paris en Compagnie
Activities: Karate (14 yrs; French Championship runner-up 2017), Surfing, Tennis, Piano